

x402, AP2 and A2A — what is actually built

Akif Kadaraca
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Pragma.Vision **DELIVERED LATE — ASSESSED BY BUILDING**

Agentic payments are usually discussed as one thing. They are three, and the separation is the whole design. *This note assesses each layer by running it rather than by reading about it — the settlement layer was proven end-to-end on Base Sepolia before this was written.*

A2A — Discovery
Who is this agent and what does it sell?
EDUCATION STAGE

AP2 — Authorization
Is this agent allowed to spend this?
CONVENTION, NOT STANDARD

x402 — Settlement
Did the money actually move?
COMMERCIAL STAGE

WHY THE LAYERS ARE SEPARATE

and why that matters commercially

NOT SOLVED BY ANYONE

the actual open problem

Each layer can be adopted without the others. A company can serve an Agent Card without accepting payments, or gate an endpoint with x402 without ever touching A2A. That lowers the cost of entry to a single afternoon, and it is the clearest explanation for why settlement is spreading faster than the layers above it.

The a2a - x402 extension is the seam between them. A buyer discovers a seller through its Agent Card, the two exchange an A2A task, the buyer carries an AP2-style mandate stating what it may spend, and settlement happens over x402. In today's tooling only the settlement half is production-shaped. The mandate half is mostly convention — everyone building now is inventing their own signed JSON and hoping it resembles whatever the ecosystem eventually adopts.

MATURE ENOUGH TO BUILD ON

verified by running it

- **The x402 exact scheme on EVM.** It rests on EIP-3009 `transferWithAuthorization`, a settled standard. The buyer signs; the facilitator submits and pays gas. My buyer wallet held 0.0001 ETH and the payment still settled — the gasless property is real, not marketing.
- **Public facilitators.** `x402.org/facilitator` handled verify and settle on Base Sepolia with no API key and no registration.
- **Middleware.** `@x402/hono` gated a route in roughly twenty lines of code.

ROUGH EDGES

where a newcomer loses a day

- **Two package generations coexist.** The legacy family uses `network: "base-sepolia"`; the current one uses `CAIP-2 eip155:84532`. Vendor documentation is split between them and mixing them fails without a useful error. This is the single most likely thing to block a newcomer and it is not flagged anywhere as a warning.
- **Cloudflare Workers needs lazy construction.** The payment middleware has to be built inside a request handler rather than at module scope. None of the official examples mention this.
- **No adopted mandate schema.** The authorization layer has no interoperable shape yet.

Settlement answers *did the money move*. Nothing in the stack answers *should this counterparty have been trusted, what happens on dispute, or who is liable when an autonomous buyer overspends inside its own limits*. This gap sits above both A2A and x402. It has research attention and no implementation. A team entering the space now is early on trust and roughly on time on settlement — which is a more useful way to describe the opportunity than any adoption figure.

EVIDENCE FROM MY OWN BUILD

rather than from documentation

`GET /reading` on a Cloudflare Worker returns `402 Payment Required` with the requirements encoded in a `PAYMENT-REQUIRED` header. A Node buyer agent decoded them, signed an EIP-3009 authorization, and retried. The facilitator settled on Base Sepolia and the Worker returned the sensor reading plus a receipt.

FIRST SETTLEMENT
`0xe18db4768d05030511485080ad850270b49e8a00df7965470a27ae2b93f4d1f3`
Base Sepolia · `eip155:84532` · 0.001 USDC · transcript in `docs/402-transcript.txt`

One observation worth reporting: the same URL behaves differently by client. A browser receives a human-readable paywall page; an agent receives machine-readable requirements. That content negotiation is a small design detail with a large implication — one endpoint can serve both markets without a second integration.

WHERE CLOUDFLARE FITS

platform assessment

Workers is a natural seller runtime: the payment gate is middleware, the free tier covers a demo indefinitely, and edge deployment means a device-facing API has no origin server to operate. Durable Objects give each simulated device a single-instance home owning its own telemetry and payment state — precisely the shape of a device twin. KV covers receipts, idempotency keys and rate-limit counters.

Cloudflare has also signalled commercial intent here with its Monetization Gateway and Pay Per Crawl, which suggests the platform bet is not incidental.

SIGNAL	TYPE	READING
x402 Foundation formed under the Linux Foundation (-40 members)	Governance	Strongest maturity signal in the set — neutral stewardship rather than one vendor’s protocol.
Cloudflare Monetization Gateway waitlist opens	Platform	Infrastructure vendors are treating agent payments as a product line, not an experiment.
DeepLearning.AI A2A course with Google Cloud and IBM	Education	A2A is still being taught rather than deployed. Courses precede adoption by roughly a year.
arXiv paper on the agentic governance gap	Research	Independent confirmation of the trust gap described on page 1.
RoboPay Agent Skill released	Tooling	Payment capability packaged for agent frameworks — the integration surface is forming.
Aethir Claw A2A payment test	Adoption	An early end-to-end trial outside the core ecosystem.
Mezo MUSD settlement rail	Settlement	Competition among settlement assets is beginning; USDC is not guaranteed to stay the default.
Reported spending volume on Base via agent payments	Adoption	Self-reported by an interested party. Cite as a claim, never as a measured figure.

Reading of the set. A2A sits at the education stage; x402 sits at the commercial stage; the governance layer above both has research attention and no implementation. Source URLs are attached separately and must be present before this note is circulated outside the team — an undated, unsourced signal is worth less than no signal at all.

A · Autonomous sensor-data market

RECOMMENDED

A simulated device agent sells individual readings — temperature, air quality — at a fixed price per call. The buyer is a monitoring or forecasting agent that needs a specific reading now and has no contractual relationship with the device owner. The seller is the device twin. Simulated: the sensor hardware and its telemetry. Real: discovery, payment authorization, settlement and receipt.

I recommend this one for an evidential rather than an aesthetic reason. The settlement path is already proven on-chain, so every claim in the one-pager can point at a transaction hash instead of an intention. It is also the most legible on a demo page: a stranger sees a number arrive and a payment settle in the same second.

Price unit \$/reading Simulated device only Status settlement proven

B · Pay-per-inference edge compute

WEEK 5 ADDITION

A Workers AI endpoint — a small classification or embedding model — behind the same 402 gate at its own price. The buyer is an agent that needs one inference and does not want to hold an API key or a billing relationship. Nothing here is simulated except the value of the tokens: the inference is real.

This is closer to real revenue logic, since it is how soft.house could eventually meter its own APIs. It is planned as the second resource type behind the same gate in Week 5, which makes it a sequencing decision rather than a competing option.

Price unit \$/inference Simulated nothing Status planned W5

C · Metered actuation and device-time rental

DEFERRED DELIBERATELY

A buyer agent rents exclusive control of a device for a bounded window — a camera pointing, a pump running, a slot on a shared machine. The buyer is a scheduling agent; the seller is a device twin holding a lease. Simulated: the actuator and its physical effect. Real: the payment and the lease accounting.

This is the most commercially interesting of the three, because time-bounded exclusivity is genuinely hard to sell without an intermediary. It is set aside for exactly that reason: it exposes the questions the other two avoid — what happens when payment succeeds but the device is already leased, and who refunds an unusable lease. Those need the mandate and dispute layers that do not exist yet, which makes this the strongest argument for continued work rather than a candidate for a nine-week sprint.

Price unit \$/minute Simulated actuator Status recorded, not built

Recommendation. Proceed with A, add B in Week 5 as a second resource type behind the same gate, and keep C on record as the case for work beyond this program. The layer that decides whether any of this becomes a business is not settlement, which now works, but the authorization and trust layers above it — and those are still open to anyone willing to implement rather than describe them.